

Do Visual Emotion Models Reason with the Cues They Report? A Counterfactual Audit of Cue-Aware Affective Reasoning

Motivation. Visual emotion systems increasingly accompany categorical or dimensional predictions with saliency, cue weights, or natural-language evidence. Yet agreement with image-level annotations does not establish that these reports describe the evidence controlling the output. A system may predict sadness from dark colour or background correlations while naming a face as justification. Recent VLM benchmarks similarly report intensity-calibration errors and shallow affective descriptions [1], but output quality alone cannot establish evidence use. We therefore ask: *do the cues a model reports agree with the cues to which its prediction is behaviourally sensitive?*

Counterfactual Affective Twins. We treat cue-aware affective reasoning as a testable contract: a system reports an emotion, valence–arousal, uncertainty, and the visible cue supporting its judgement. We audit that contract with controlled original–counterfactual pairs. Each “twin” changes one inspectable cue family—an AU-related facial region, colour and lighting, scene context, or embedded text—and records the edit mask and preserved content. A system passes only when four behaviours agree: it responds to the changed evidence, remains invariant to preserved content, grounds its report in the manipulated region, and reduces confidence when cues genuinely conflict. These tests separate *cue responsiveness*, *selective invariance*, *evidence grounding*, and *conflict calibration*, which ordinary accuracy conflates. The same protocol applies to learned cue routers and VLM-generated explanations.

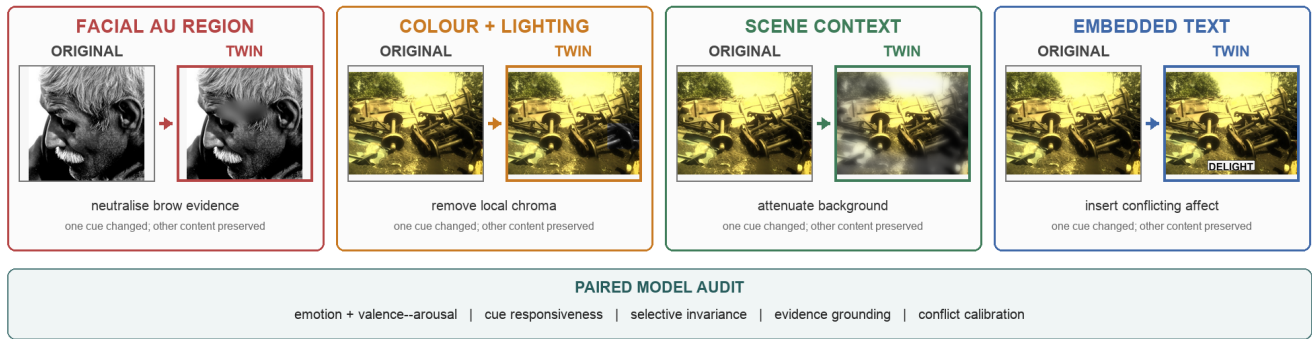


Figure 1: Representative outputs from the implemented intervention generator. Quantitative evidence is reported for colour/lighting; AU-related face, context, and text pairs illustrate the full audit design pending human validation.

Feasibility study. We first test a necessary safeguard: does an affect-sensitive intervention selected by one model transfer to an independently fitted evaluator, or merely exploit the locator? Emotion6 contains 1,980 images across six categories and retains viewer-response distributions [2]. Using frozen ImageNet-pretrained ResNet-18 features [3], we train two emotion heads with independent initialisation and bootstrap samples; the evaluator obtains 51.7% accuracy and 0.512 macro-F1 on the 402-image test split. For 60 balanced held-out images, the locator ranks 16 equal-area regions by the reduction in annotated-class probability after luminance-preserving chroma removal. The evaluator then assesses the selected region against all 15 area-matched controls. Selected interventions reduce annotated-class probability by 0.070, compared with -0.001 for controls. The paired advantage is 0.071 (95% bootstrap CI [0.050, 0.096]) and is positive for 58 of 60 images. Thus, the detected colour reliance transfers across independently fitted heads rather than remaining a locator-specific saliency artefact.

From feasibility to benchmark. The executable prototype now produces twins, masks, matched controls, and intervention records for all four cue families. Human raters will next verify edit naturalness, target-cue change, and non-target preservation before the pairs are treated as perceptual counterfactuals. We will then compare a weakly supervised cue-routing predictor, conventional emotion classifiers, and VLMs. The primary analysis contrasts *reported reliance*—router weights or named evidence—with *behavioural reliance*, measured by target-versus-control sensitivity. This report–response alignment exposes which evidence a model uses, ignores, or merely mentions, and whether uncertainty rises under genuine cue conflict. The resulting audit connects facial behaviour, dimensional affect, scene emotion, and emotion-aware captioning through a common requirement: affective evidence should be not only reported, but independently testable.

References. [1] D. She et al. “AICA-Bench: Holistically Examining the Capabilities of VLMs in Affective Image Content Analysis.” *Findings of ACL*, 2026. [2] K.-C. Peng et al. “A Mixed Bag of Emotions: Model, Predict, and Transfer Emotion Distributions.” *CVPR*, 2015. [3] K. He et al. “Deep Residual Learning for Image Recognition.” *CVPR*, 2016.